

section 5.2, 5.4 - Linear Independence, Basis & Dimensions

Linear Independence:

A set of vectors is linearly independent if & only if...

$$c_1 x_1 + c_2 x_2 + \dots + c_n x_n = 0$$

$$\Rightarrow \text{where } c_1 = c_2 = c_3 = \dots = c_n = 0$$

* $\rightarrow$ 2 vectors $\Rightarrow$ dependent if parallel

$\rightarrow$ 3 vectors $\Rightarrow$ dependent if one is linear combination of other two

$\rightarrow$ Any vector expressed from others $\Rightarrow$ dependent

e.g. verify $S = \left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}$ is linearly independent

subset of $\mathbb{R}^4$.

$$c_1 x_1 + c_2 x_2 + c_3 x_3 = 0$$

$$c_1 \begin{bmatrix} 1 \\ 1 \\ 2 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 1 \\ 1 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} 3 \\ 0 \\ 1 \\ 1 \end{bmatrix} = 0$$

$$\begin{bmatrix} c_1 + 2c_2 + 3c_3 \\ c_1 + c_2 \\ 2c_1 + c_2 + c_3 \\ 0 \quad c_2 + c_3 \end{bmatrix} \Rightarrow \begin{cases} c_1 + 2c_2 + 3c_3 = 0 \\ c_1 + c_2 = 0 \\ 2c_1 + c_2 + c_3 = 0 \\ c_2 + c_3 = 0 \end{cases}$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 1 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$c_1 = c_2 = c_3 = 0 \Rightarrow$ Thus linearly independent

e.g.

check if linearly independent or dependent

1. $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\} \Rightarrow$ linearly independent

2. $\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} \right\} \Rightarrow$ can be $\begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} = (2) \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ parallel
linearly dependent

3. $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\} \Rightarrow$ can be $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, linear combo
linearly dependent

4. $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \Rightarrow$ linearly independent

Basis :

A basis, B , of a subspace U is a set B s.t. ...

① B spans U

② B is linearly independent

$\Rightarrow B = c_1 x_1 + c_2 x_2 + \dots + c_n x_n$, minimal spanning set

Dimension of Basis :

The dimension of a subspace = # of vectors in Basis

$\Rightarrow$ Line $\rightarrow \dim = 1$

$\mathbb{R}^n \rightarrow \dim = n$

Plane $\rightarrow \dim = 1$

Row & Column of A :

for matrix A that is $m \times n$, $A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$

- $\text{row}(A) = \{[a \ b \ c], [d \ e \ f]\}$, $\mathbb{R}^n$

- $\text{col}(A) = \left\{ \begin{bmatrix} a \\ d \end{bmatrix}, \begin{bmatrix} b \\ e \end{bmatrix}, \begin{bmatrix} c \\ f \end{bmatrix} \right\}$, $\mathbb{R}^m$

Fundamental Subspace of a Matrix :

for $m \times n$ matrix A ...

- Basis of $\text{row}(A) = \{ \text{non-zero rows of RREF} \}$

- Basis of $\text{col}(A) = \text{im}(A) = \{ \text{columns of A, leading 1's in RREF} \}$

- Basis of $\text{null}(A) = \{ \text{basic solutions} \}$, $\dim = n - r$

e.g.

let $A = \begin{bmatrix} 1 & 0 & 1 & -1 \\ 3 & 0 & 1 & -1 \\ 5 & 0 & 3 & -1 \end{bmatrix}$, find basis for $\text{im}(A)$, $\text{row}(A)$,
 $\text{col}(A) = \text{im}(A)$, $\text{null}(A)$

$\text{im}(A)$ & $\text{col}(A)$ & $\text{row}(A)$:

$$R = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Leftarrow A = \begin{bmatrix} 1 & 0 & 1 & -1 \\ 3 & 0 & 1 & -1 \\ 5 & 0 & 3 & -1 \end{bmatrix}$$

$\underbrace{\quad}_{x_1} \quad \underbrace{\quad}_{x_3}$ $\underbrace{\quad}_{x_1} \quad \underbrace{\quad}_{x_3}$

corresponding columns ...

$$\text{Col}(A) = \text{im}(A) = \left\{ \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} \right\}$$

$$\text{row}(A) = \left\{ [1 \ 0 \ 0 \ 1], [1 \ 0 \ 1 \ -2] \right\}$$

null(A):

$$R = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{matrix} x_1 = -t \\ x_2 = s \\ x_3 = 2t \\ x_4 = t \end{matrix} \Rightarrow x = s \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 2 \\ 1 \end{bmatrix}$$

$$\text{null}(A) = \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 2 \\ 1 \end{bmatrix} \right\}$$

Rank & linear dependency:

given $\text{rank}(A) = r$ and A is $m \times n$ matrix ...

if ... $r < \# \text{ of columns} \Rightarrow$ linearly dependent

$r = \# \text{ of columns} \Rightarrow$ linearly independent

Invertible Matrix Theorem ($n \times n$):

If one is true $\Rightarrow$ All are true, if one is false $\Rightarrow$ all are false ...

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|---|--|---|
| <ul style="list-style-type: none"> • $\det(A) \neq 0$ • A is invertible • Column are indep. • Columns span $\mathbb{R}^n$ • columns form B for $\mathbb{R}^n$ | | <ul style="list-style-type: none"> • row are indep. • rows span $\mathbb{R}^n$ • rows form B for $\mathbb{R}^n$ |
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